

EVALUATION OF SUPERCLASSICAL THERMODYNAMICS

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Beyond Quantum

April of 2026

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1. GOOD POINTS

1.1. Emergent Nature and Explanation

This theory argues that quantum mechanics is not a fundamental theory, but rather that there is something underlying it, which would be the thermodynamic zero-point field. It manages to explain wave-particle duality much more appropriately than Bohmian mechanics already did, and it solves the measurement problem.

What distinguishes it positively from Bohmian mechanics is that it provides an origin for the quantum potential without the need to invent new physics, but rather as an emergent phenomenon from physics we already know well, without the need to postulate anything new.

In this way, the quantum potential would not be a new force but the result of accumulated kinetic energy (heat). Trajectories curve by bouncing against the density gradients of this physical heat accumulated in the vacuum.

(Here it would be necessary to see what exactly this zero-point field is, whether it is truly something new or not, and what its characteristics are.)

1.2. A 3D Space

Bohmian mechanics requires an abstract “pilot wave” moving in a multidimensional space, which seems like a “mathematical artifice” just as Bohm himself viewed it. This theory manages to derive all trajectories and interactions purely using standard three-dimensional space and the superposition of standing waves.

1.3. Testable Predictions (The Collapse of Quantum Laws)

The theory is not just a mathematical reinterpretation; it predicts new physics. It states that if we increase the temporal resolution of our experiments to observe at time scales on the order of the particle’s agitation motion ($t \simeq 1/\omega$), the thermodynamic averages break down and the ordinary laws of quantum mechanics (along with the Born Rule) would collapse completely, revealing a purely chaotic, irreversible environment governed by the stochastic physics of classical fluids.

2. BAD POINTS

2.1. Non-Locality and Entanglement

The theory is consistent and coherent and manages to predict both phenomena; however, it has some major flaws. On the one hand, being a deterministic and non-local theory, it clashes head-on with special relativity, just as Bohm's theory does. Although it proposes some solutions, none are developed nor possible at the moment.

On the other hand, to achieve this non-locality, it makes use of the heat diffusion equation. This equation, which is used in classical physics, is an imperfect model because it is non-local while classical physics is local. Therefore, a good question is: why can we use it now when before it was only a model? The answer given is that perhaps this is not the definitive equation, but rather there would be another one that is indeed superluminal but not instantaneous; however, this has not yet been explored.

It's in these points where the major shortcomings of the theory are found.

3. NEUTRAL POINTS

3.1. Indeterminism: Ontological vs Epistemic

Heisenberg’s uncertainty principle emerges directly from our inability to measure without changing the underlying thermal equilibrium. The particle has a well-defined position and momentum, unlike in orthodox quantum mechanics. The vacuum noise causes an indeterminacy that matches Heisenberg’s exactly through the Fisher Length, and it will always be greater than this due to the Cramér-Rao bound, since by attempting to measure, we also apply more thermal noise.

3.2. Continuous Energy Exchange and Emergent Quantization

Unlike orthodox quantum mechanics, this theory posits that energy is not fundamentally discrete. The underlying physical reality consists of a continuous and dynamic exchange of energy (heat) between the oscillating particle and the zero-point field. Because the particle operates as a dissipative system in a non-equilibrium steady state, it must continuously absorb and emit thermal energy to sustain its oscillation.

The discreteness we observe in experiments—quantization according to the Planck-Einstein relation $E = \hbar\omega$ —is merely an emergent, macroscopic phenomenon. It represents the exact statistical average of this continuous heat exchange over one full oscillation cycle. The authors refer to this discrete average as a “fleeting constancy”, proving that the fundamental thermodynamic processes of nature remain continuous, and quantization is just the balanced macroscopic average required for the particle’s stability.

3.3. Multiple Steady States: Physical Decay vs. Probabilistic Cancellation

Orthodox quantum mechanics (such as Feynman’s path integral formulation) assumes that all possible paths exist, but those that do not minimize the action mathematically cancel each other out through destructive probability interference. The superclassical theory replaces this abstract mathematical cancellation with real thermodynamic instability and physical orbit decay.

When a particle moves in a confining potential, it continuously interacts with its own “path memory”—the wake of thermal diffusion waves it previously emitted into the vacuum. The particle can physically attempt to follow any continuous, arbitrary orbit, but those that do not perfectly resonate with its wave wake disrupt the delicate energetic balance.

In these non-resonant orbits, the particle dissipates more heat into the vacuum than it absorbs, causing its trajectory to become thermodynamically unstable. Therefore, these “forbidden” orbits actually occur, but they rapidly decay. The particle loses momentum and alters its path until it “self-traps” into one of the few resonant, stable configurations—the multiple steady states. In these specific quantized orbits, the continuous heat exchange perfectly balances out, allowing the particle to remain there indefinitely without further decay.